

Bank Customer Churn Analysis

Exploratory Data Analysis — uncovering why customers leave and who is most at risk

EDA PROJECT

KAGGLE DATASET



Project Scope

What We Set Out to Understand



The Question

Why do bank customers leave — and which segments are most at risk?

The Method

Segment-level groupby analysis, correlation heatmaps, and visual EDA

The Goal

Quantify churn risk by segment to inform targeted retention campaigns

Dataset

10,000 Customers. 14 Variables.

10K

Customer Records

14

Feature Columns

20.4%

Overall Churn Rate

3

Tools Stack

Python · Pandas · Matplotlib · Seaborn

Source: Churn Modelling Dataset (Kaggle) — includes demographics, product holdings, account activity, and exit status.



From Raw Data to Actionable Insight

Explore Visually
Plot charts across age,
geography, and activity.

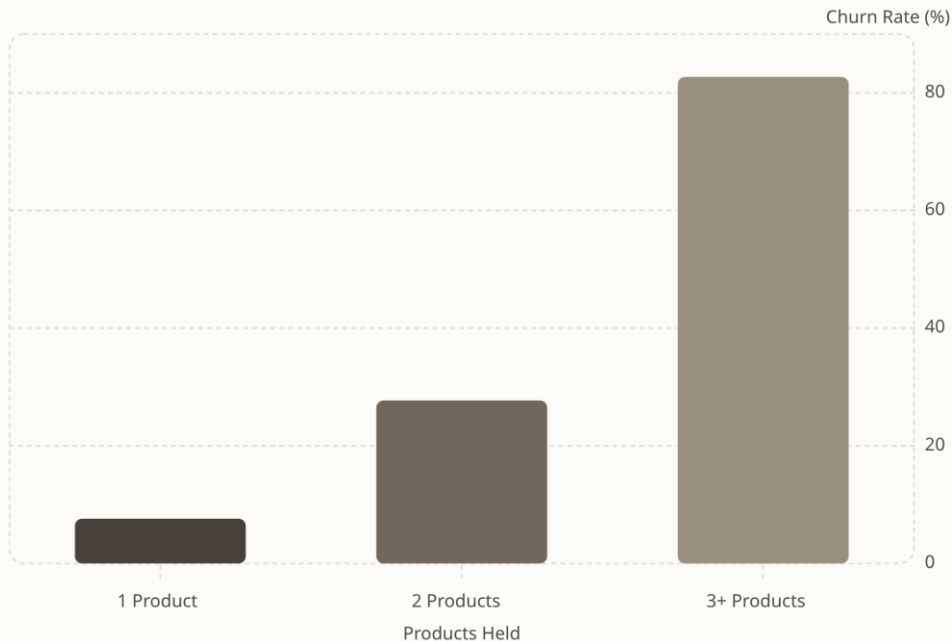
Segment Analysis
Group by segments and
compute churn rates.

Load & Clean
Remove irrelevant columns
and handle missing values.

Validate
Use correlation heatmap to
confirm segment findings.

Key Finding #1

Product Count: The Strongest Churn Driver

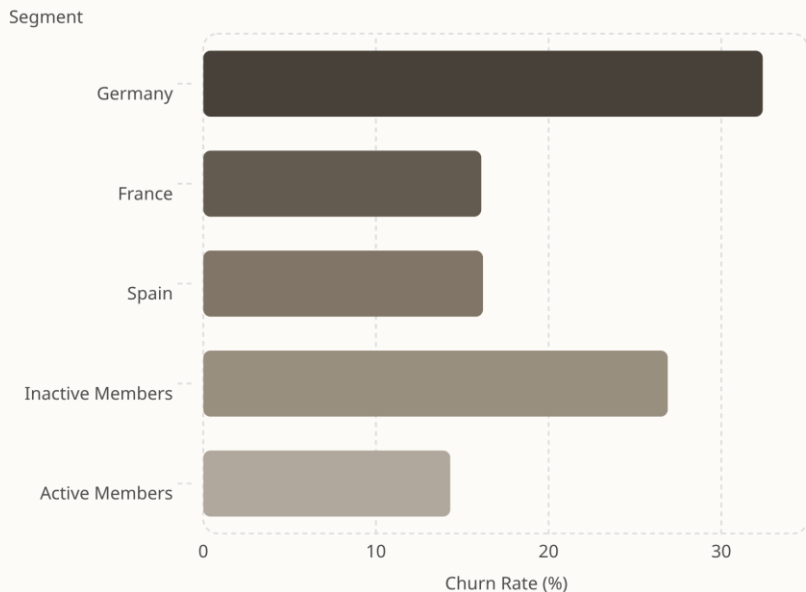


Non-linear pattern: Correlation heatmap shows ~0 for NumOfProducts–Exited. Segment-level analysis was required to catch this.

Customers with 3+ products churn at **82.7%** (n=266) — and 100% for the 60 customers holding all products.

Key Finding #2

Geography & Activity Status



Geography

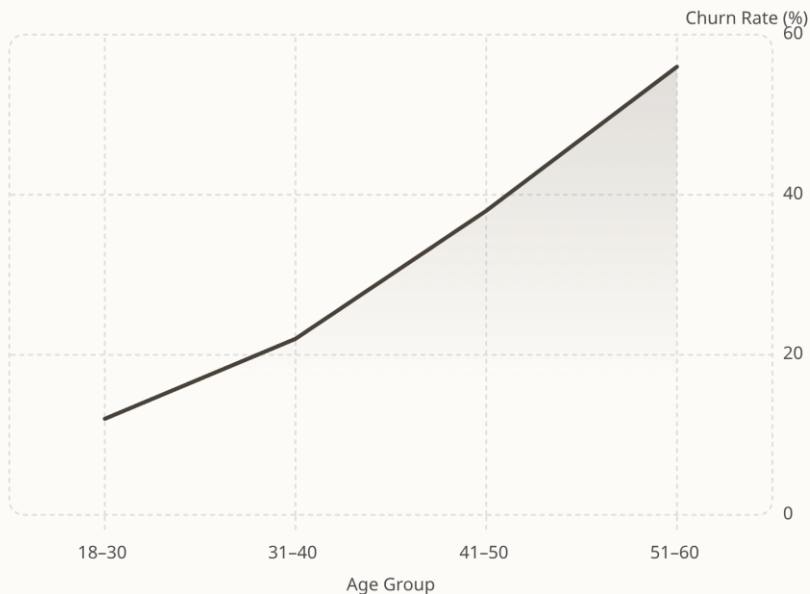
Germany churns at **32.4%** — roughly double France and Spain (~16%).

Activity Status

Inactive members churn at **26.9%** vs. **14.3%** for active members — nearly 2× the risk.

Key Finding #3

Age, Gender & Credit Score



Age

Churn peaks at **56%** for ages 51–60. Age has the strongest linear correlation ($r = 0.30$).

Gender

Female customers churn at **25.1%** vs. **16.5%** for males.

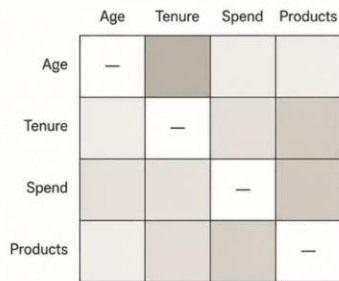
Credit Score

Correlation ~ 0.0 — **no predictive power** for churn in this dataset.

When Correlation Misleads

Correlation Heatmap

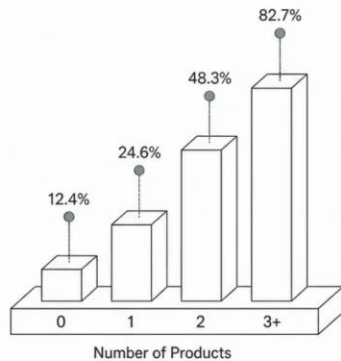
(linear relationships only,
misses non-linear patterns
like product count)



Shows Age $r=0.30$
as strongest

Segment-Level Groupby

(captures non-linear churn spikes,
reveals 3+ products = 82.7% churn,
validates small sample sizes)



Reveals 3+ products =
82.7% churn

Validates small
sample sizes

The Lesson

A correlation heatmap alone would have missed the strongest churn driver entirely.

- NumOfProducts-Exited correlation ≈ 0
- Segment analysis reveals 82.7% $\rightarrow$ 100% churn
- Always validate correlations with segment-level breakdowns

Actionable Output

Priority Retention Segments



3+ Product Holders

82.7–100% churn rate. Highest-risk segment by far.



Germany-Based Customers

32.4% churn — double other geographies. Investigate regional drivers.



Inactive Members

26.9% churn vs. 14.3% active. Re-engagement campaigns needed.



Customers Ages 51–60

56% churn rate. Age is the strongest linear predictor.



Key Takeaways

What This Analysis Revealed



Segment-level analysis is non-negotiable

Correlation alone missed the strongest churn driver.



Non-linear patterns require groupby, not just heatmaps

Product count's effect is invisible in linear correlation.



Sample size matters

Small segments (n=60 for 3+ products) need cautious interpretation.



Bottom line: Target 3+ product holders, inactive members, and Germany-based customers first for maximum retention impact.